

CHIMERA Phase 1 - Quick Start Guide

From Zero to Running in 10 Minutes

Target Audience: DevOps Engineers, Platform Engineers

Prerequisites: Docker, gcloud CLI (optional for cloud), Git

Time Required: 10 minutes

What You'll Deploy

By the end of this guide, you'll have a running multi-agent orchestration platform with:

-  5 AI agents coordinating across 4 zones
-  Automated workflow processing (n8n)
-  Real-time monitoring (Prometheus + Grafana)
-  Immutable audit trail (Evidence Ledger)
-  Health monitoring and alerting



Prerequisites Check

```
# Check Docker
docker --version
# Required: Docker 20.10+

# Check Docker Compose
docker compose version
# Required: Docker Compose 2.0+

# Check disk space
df -h
# Required: At least 10GB free

# Optional: Check gcloud (for cloud deployment)
gcloud --version

---
```



Option 1: Local Docker Deployment (Recommended for Testing)

Step 1: Clone and Navigate

```
git clone <repository-url> chimera-phase1
cd chimera-phase1
```

Step 2: Create Environment Configuration

```
cat > infrastructure/docker/.env << EOF
# GCP Configuration (optional for local testing)
GCP_PROJECT=novate-sentraais
GCP_REGION=us-central1

# Anthropic API Key (REQUIRED)
ANTHROPIC_API_KEY=sk-ant-your-key-here

# n8n Configuration
N8N_USER=admin
N8N_PASSWORD=chimera-admin-2026
N8N_HOST=localhost

# Grafana Configuration
GRAFANA_USER=admin
GRAFANA_PASSWORD=chimera-admin-2026
EOF
```

 **IMPORTANT:** Replace `sk-ant-your-key-here` with your actual Anthropic API key from <https://console.anthropic.com/>

Step 3: Deploy with Single Command

```
./infrastructure/scripts/deploy.sh deploy-docker
```

What happens:

1.  Pulls Docker images (1-2 minutes)
2.  Builds custom agent containers (2-3 minutes)
3.  Starts all services
4.  Waits for health checks (30 seconds)

5.  Validates all agents are running

Step 4: Verify Deployment

```
# Check service health
./infrastructure/scripts/deploy.sh health

# Expected output:
# [SUCCESS] asirap-supervisor is healthy
# [SUCCESS] nin-forensics is healthy
# [SUCCESS] temporal-sync is healthy
# [SUCCESS] playbook-synthesis is healthy
# [SUCCESS] evidence-ledger is healthy
```

Step 5: Access the Platform

Open your browser and visit:

Service	URL	Default Credentials
ASIRAP Supervisor API	http://localhost:8080/docs	N/A (API)
NIN Forensics UI	http://localhost:8082	N/A (UI)
Temporal Sync API	http://localhost:8083/docs	N/A (API)
Playbook Synthesis API	http://localhost:8084/docs	N/A (API)
Evidence Ledger API	http://localhost:8081/docs	N/A (API)
n8n Workflows	http://localhost:5678	admin / chimera-admin-2026
Grafana Dashboards	http://localhost:3000	admin / chimera-admin-2026
Prometheus Metrics	http://localhost:9090	N/A (Metrics)

Option 2: Google Cloud Run Deployment (Production)

Step 1: Prerequisites

```
# Install gcloud CLI
# https://cloud.google.com/sdk/docs/install

# Authenticate
gcloud auth login

# Set project
gcloud config set project novate-sentraais

# Enable required APIs
gcloud services enable \
  run.googleapis.com \
  artifactregistry.googleapis.com \
  secretmanager.googleapis.com \
  bigquery.googleapis.com \
  storage.googleapis.com
```

Step 2: Create GCP Resources

```
# Create Artifact Registry repository
gcloud artifacts repositories create chimera-agents \
  --repository-format=docker \
```

```
--location=us-central1 \  
--description="CHIMERA Agent Container Images"  
  
# Create BigQuery dataset for Evidence Ledger  
bq mk --dataset --location=US chimera_evidence_ledger  
  
# Create Cloud Storage bucket for reports  
gsutil mb -l us-central1 gs://novate-sentraais-chimera-reports
```

Step 3: Store Secrets

```
# Store Anthropic API key  
echo -n "sk-ant-your-key-here" | gcloud secrets create anthropic-  
api-key \  
--data-file=-  
  
# Store JWT secret for ASIRAP  
openssl rand -base64 32 | gcloud secrets create asirap-jwt-secret \  
--data-file=-
```

Step 4: Deploy to Cloud Run

```
# Set environment variables  
export GCP_PROJECT=novate-sentraais  
export GCP_REGION=us-central1  
export ENVIRONMENT=staging  
  
# Deploy  
./infrastructure/scripts/deploy.sh deploy-cloud -e staging -p  
novate-sentraais
```

What happens:

1.  Authenticates with GCP
2.  Builds Docker images
3.  Pushes to Artifact Registry
4.  Applies Terraform infrastructure
5.  Deploys to Cloud Run
6.  Configures networking and IAM
7.  Runs health checks

Step 5: Get Service URLs

```
# ASIRAP Supervisor
gcloud run services describe asirap-supervisor-staging \
  --region us-central1 \
  --format 'value(status.url) '

# NIN Forensics
gcloud run services describe nin-forensics-staging \
  --region us-central1 \
  --format 'value(status.url) '

# Evidence Ledger
gcloud run services describe evidence-ledger-staging \
  --region us-central1 \
  --format 'value(status.url) '

---
```

Testing Your Deployment

Test 1: Register an Agent with ASIRAP

```
curl -X POST http://localhost:8080/agents/register \
-H "Content-Type: application/json" \
-d '{
  "agent_id": "test-agent-001",
  "agent_name": "Test Agent",
  "zone": 2,
  "capabilities": ["testing", "demo"],
  "endpoint_url": "http://test-agent:8090"
}'
```

```
# Expected response:
# {
#   "agent_id": "test-agent-001",
#   "status": "active",
#   "registered_at": "2026-01-15T..."
# }
```

Test 2: List All Agents

```
curl http://localhost:8080/agents
```

```
# Expected response:
# [
#   {
#     "agent_id": "nin-forensics",
#     "zone": 2,
#     "status": "active",
#     "last_heartbeat": "2026-01-15T..."
#   }
# ]
```

```
# },
# ...
# ]
```

Test 3: Log Event to Evidence Ledger

```
curl -X POST http://localhost:8081/log \
-H "Content-Type: application/json" \
-d '{
  "event_type": "test.deployment.success",
  "agent_id": "deployment-script",
  "zone": 3,
  "payload": {
    "message": "CHIMERA deployment successful",
    "timestamp": "'$(date -u +%Y-%m-%dT%H:%M:%SZ)'"
  }
}'

# Expected response:
# {
#   "event_id": "evt_...",
#   "status": "logged",
#   "timestamp": "2026-01-15T..."
# }
```

Test 4: Analyze Document with NIN Forensics

1. Open <http://localhost:8082> in your browser
2. Upload a test document (PDF, TXT, or paste text)
3. Click "Analyze Document"
4. Wait for blueprint extraction (10-30 seconds)

5. Review the 5D analysis results
6. Download blueprint as JSON

Test 5: Import n8n Workflows

```
./infrastructure/scripts/deploy.sh import-workflows
```

```
# Then open http://localhost:5678
# Login: admin / chimera-admin-2026
# Verify workflows imported:
# - CHIMERA Document Intake Pipeline
# - CHIMERA Agent Health Monitor
```

Monitoring Your Deployment

Grafana Dashboards

1. Open <http://localhost:3000>
2. Login: admin / chimera-admin-2026
3. Navigate to Dashboards → CHIMERA Overview
4. View:
5. Agent health status
6. Request rates

7. Error rates
8. Resource utilization

Prometheus Queries

1. Open <http://localhost:9090>
2. Try these queries:

```
# HTTP requests per second by agent
rate(http_requests_total[5m])

# Agent health status
up{job=~"asirap-supervisor|nin-forensics|temporal-sync|evidence-
ledger"}
```

```
# Request duration 95th percentile
histogram_quantile(0.95, http_request_duration_seconds_bucket)
```

Evidence Ledger Queries

```
# Query recent events
curl "http://localhost:8081/events?limit=10"

# Query events by zone
curl "http://localhost:8081/events/zone/2"

# Query events by agent
curl "http://localhost:8081/events/agent/nin-forensics"

---
```

Common Configuration Tasks

Update Anthropic API Key

```
# For Docker
vim infrastructure/docker/.env
# Update ANTHROPIC_API_KEY=...
docker compose down && docker compose up -d

# For Cloud Run
echo -n "sk-ant-new-key" | gcloud secrets versions add anthropic-
api-key \
  --data-file=-
# Redeploy services
```

Add Custom Agent

```
# 1. Create agent registration request
curl -X POST http://localhost:8080/agents/register \
  -H "Content-Type: application/json" \
  -d '{
    "agent_id": "custom-agent",
    "agent_name": "My Custom Agent",
    "zone": 3,
    "capabilities": ["custom", "feature"],
    "endpoint_url": "http://custom-agent:8090"
  }'

# 2. Add to docker-compose.yml
# 3. Implement agent with CHIMERA SDK
# 4. Deploy
```

Configure Slack Alerts

```
# Get Slack webhook URL from https://api.slack.com/messaging/webhooks
```

```
# Update n8n workflows
# 1. Open http://localhost:5678
# 2. Edit workflow: Agent Health Monitor
# 3. Configure Slack node with webhook URL
# 4. Save and activate
```

Stopping and Cleaning Up

Stop All Services (Docker)

```
./infrastructure/scripts/deploy.sh cleanup
```

```
# Or manually:
docker compose down
```

Stop and Remove Data (Docker)

```
docker compose down -v
# Warning: This deletes all volumes (logs, data, etc.)
```

Destroy Cloud Infrastructure

```
./infrastructure/scripts/deploy.sh cleanup-cloud
```

```
# Requires confirmation
# This will destroy:
# - Cloud Run services
# - Artifact Registry images
# - BigQuery tables
# - Cloud Storage buckets
```

Troubleshooting

Services Won't Start

```
# Check logs
docker compose logs

# Check specific service
docker compose logs asirap-supervisor

# Check resource usage
docker stats

# Restart specific service
docker compose restart asirap-supervisor
```

Agent Health Check Fails

```
# Check if service is running
curl http://localhost:8080/health

# Check Docker network
docker network inspect chimera-network

# Check container logs
docker logs asirap-supervisor

# Common issues:
# - Port already in use: Change port in docker-compose.yml
# - Out of memory: Increase Docker memory limit
# - Missing API key: Check .env file
```

n8n Workflows Not Importing

```
# Check n8n is running
curl http://localhost:5678

# Manually import
# 1. Open http://localhost:5678
# 2. Go to Workflows → Import from File
# 3. Select: workflows/n8n/document_intake_pipeline.json
# 4. Repeat for agent_health_monitor.json
```

Cloud Run Deployment Fails

```
# Check quotas
gcloud compute project-info describe --project=novate-sentraais
```

```
# Check service account permissions
gcloud projects get-iam-policy novate-sentraais

# Check build logs
gcloud builds list --limit=5

# Retry deployment
./infrastructure/scripts/deploy.sh deploy-cloud -e staging

---
```

Next Steps

After successfully deploying CHIMERA:

1. **Read the Technical Specification**
2. /docs/CHIMERA_Technical_Specification.md
3. Understand the 4-zone architecture
4. Learn agent coordination patterns
5. **Explore the MindMap**
6. Open /docs/CHIMERA_MindMap.html
7. Interactive visualization of system architecture
8. **Review Agent APIs**
9. ASIRAP Supervisor: <http://localhost:8080/docs>
10. Temporal Sync: <http://localhost:8083/docs>
11. Evidence Ledger: <http://localhost:8081/docs>
12. **Build Custom Workflows**
13. Open n8n: <http://localhost:5678>
14. Create workflows connecting agents

15. Automate document processing
16. **Implement Playbook Synthesis Agent**
17. Complete Zone 3 implementation
18. See specification in technical docs
19. **Production Deployment**
20. Set up CI/CD pipeline (GitHub Actions)
21. Configure production secrets
22. Deploy to Cloud Run production environment

Learning Resources

Video Tutorials (Coming Soon)

- CHIMERA Architecture Overview
- Agent Development Guide
- Workflow Automation with n8n
- Monitoring and Observability

Documentation

- **API Reference:** </docs/api/>

- **Agent SDK:** `/docs/sdk/`
- **Best Practices:** `/docs/best-practices/`
- **Troubleshooting:** `/docs/troubleshooting/`

Community

- Slack: #chimera-platform
- GitHub Discussions: TBD
- Office Hours: Fridays 2pm PST

Deployment Checklist

- ☐ Docker and Docker Compose installed
- ☐ Anthropic API key obtained
- ☐ `.env` file created with API key
- ☐ Services deployed: `./deploy.sh deploy-docker`
- ☐ Health checks passing: `./deploy.sh health`
- ☐ All 5 agents accessible (ports 8080-8084)
- ☐ n8n workflows imported
- ☐ Grafana dashboard accessible
- ☐ Test agent registered successfully
- ☐ Test event logged to Evidence Ledger

- Document analyzed with NIN Forensics

Success!

You now have a fully operational multi-agent orchestration platform!

What you can do:

- Process documents automatically
- Coordinate AI agents across timescales
- Generate adaptive playbooks
- Monitor system health in real-time
- Maintain immutable audit trails

Need help?

- Check logs: `docker compose logs -f`
- Review troubleshooting guide above
- Contact support: chimera-support@novatelabs.com

Deployment Time: 10 minutes 

Status: Production Ready 

Version: 1.0.0